C 21297 **(Pages : 3)** **Name.....**

Reg. No.....

FOURTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION, APRIL 2022

Mathematics

MAT 4C 04—MATHEMATICS

(2014—2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Part A (Objective Type)

Answer all the **twelve** *questions.*
Each question carries 1 *mark.*

1. Write Laplace transform of $f''(t)$.
2. Write the general form of second order linear ODE.
3. Find L (e^{at}).
4. What is unit step function ? Give an example.
5. Give a formula for an error for Simpson's rule.
6. Find the fundamental period for sin x.
7. Find Wronskian of cos ωx and sin ωx.
8. What is particular solution of an ODE ?
9. What do you mean by an even function give example.
10. Write the 1-dimensional Heat equation.
11. State second shifting theorem for Laplace transfom.
12. Solve $y'' - y = 0$.

(12 × 1 = 12 marks)

Part B (Short Answer Type)

Answer any **nine** *questions.*
Each question carries 2 *marks.*

13. Find a basis for the solution of the differential equation $y'' - y = 0$.
14. Show that Laplace transform is a linear operator.

Turn over

15. Solve the initial value problem $y'' + 2y' + 2y = 0, y(0) = 1, y'(0) = -1$.

16. Factor $(D^2 + 6D + 13I)y = 0$ and solve it.

17. Find $L^{-1}\left(\frac{\sqrt{8}}{(s+\sqrt{2})^3}\right)$.

18. If $f(x)$ is a periodic function of x of period p, show that $f(ax), a \neq 0$, is a periodic function of x of period $\frac{p}{a}$.

19. Find the Fourier cosine transform of $e^{-ax}, a > 0$.

20. Find an ODE for the basis e^{2x}, e^{x}.

21. Solve $y'' - y = t, y(0) = 1, y'(0) = 1$ by applying Laplace transform.

22. Check whether the functions $5 \sin x \cos x$, $3 \sin 2x$, $x > 0$ are linearly independent.

23. Find solutions u of the PDE $u_{xx} - u = 0$.

24. Find an upper bound for the error incurred in estimating $\int_0^2 5x^4 dx$ using Simpson's rule with $n = 4$.

(9 × 2 = 18 marks)

Part C (Short Essays)

Answer any **six** *questions.*
Each question carries 5 *marks.*

25. Find $L^{-1}\left(\frac{1}{(s^2 + w^2)^2}\right)$.

26. Find solution of the initial value problem $y'' + 4y = 16 \cos 2x, y(0) = 0, y'(0) = 0$.

27. Find the Laplace transform of $e^{-\alpha t} \cos \beta t$.

28. Find the inverse transform $f(t)$ of $F(s) = \frac{e^{-s}}{s^2 + \pi^2} + \frac{e^{-2s}}{s^2 + \pi^2} + \frac{e^{-3s}}{(s+2)^2}$.

29. Find a general solution of the differential equation :

$y'' + 3y' + 2y = 30e^{2x}$.

30. Find the Fourier series of $f(x) = \begin{cases} -k, \text{ if } -\pi < x < 0 \\ k, \text{ if } 0 < x < \pi \end{cases}$ and $f(x + 2\pi) = f(x)$.

31. How many subdivisions should be used in the Trapezoidal Rule to approximate $\ln 2 = \int_1^2 \frac{1}{x}\,dx$ with an error whose absolute value is less than 10^{-4}.

32. Given $y' = x(1 - y), y(1) = 0, dx = 0.2$. Find the first three approximations by improved Euler method. Compare with exact solution.

33. Evaluate $\int_{-1}^{1} (1 + x^2)\,dx$ with $n = 4$ steps and find an upper bound for $|E_s|$ using Simpson's rule.

(6 × 5 = 30 marks)

Part D

Answer any **two** *questions.*
Each question carries 10 *marks.*

34. Solve $y'' + 3y' + 2y = 1$ if $0 < t < a$ and 0 if $t > a$; $y(0) = 0, y'(0) = 0$.

35. Find the Fourier series of $f(x) = x^2$ in $[-\pi, \pi]$ with $f(x + 2\pi) = f(x)$. Hence deduce that

$1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} \ldots = \frac{\pi^2}{12}$.

36. Solve the integral equation $y(t) - \int_0^t (1 + \tau)\, y(t - \tau)\, d\tau = 1 - \sinh t$.

(2 × 10 = 20 marks)